

Annotation Protocol

Computer Vision for GRIT

This annotation project is part of the GRIT project (Global Repository of Insect Traits). To compile this dataset, we want to automatically extract as many insect traits as possible from insect images.

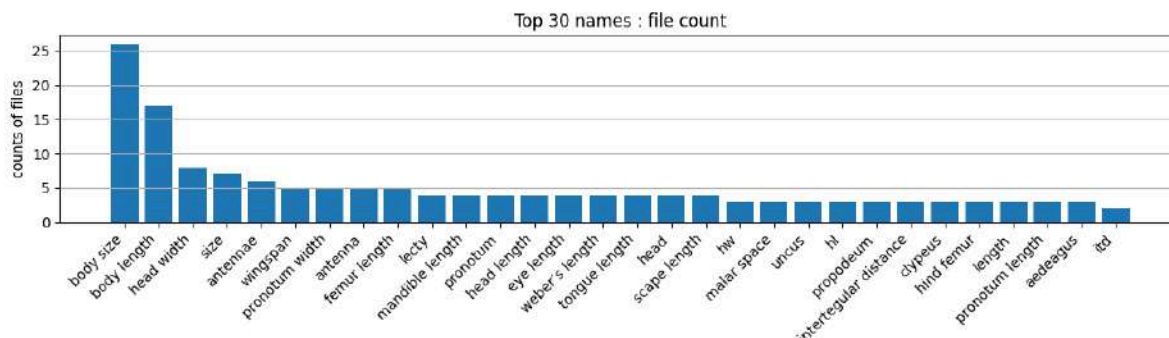
To achieve this, we need to train machine-learning models. Training these models requires a large number of annotated images. This document explains how to perform those annotations.

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Foreword

Searching for traits to extract

Based on several papers measuring traits in different kinds of insects, we identified the most common traits used in the literature.



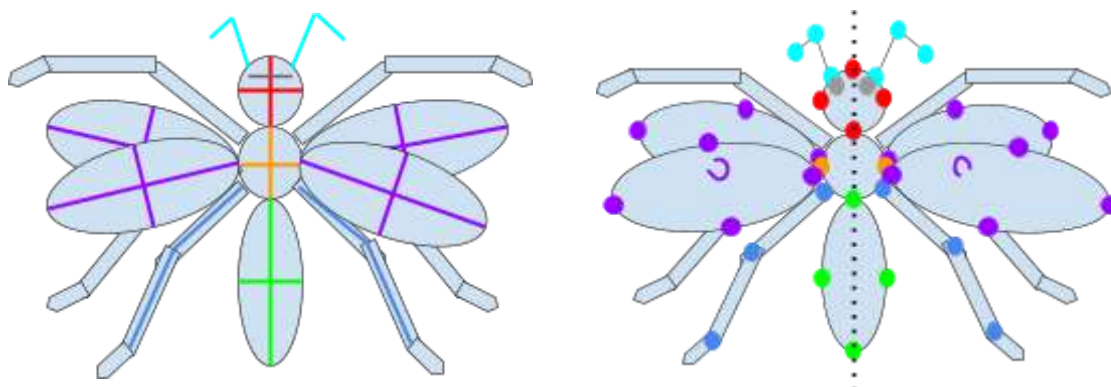
Note that some traits can be derived from others, such as total length or wing-to-body ratio. For example, the total length can be calculated as the sum of the head, prothorax, and abdomen lengths.

The selected traits to be measured are:

- head, prothorax, and abdomen widths and lengths;
- wing length and width;
- intertegular distance (distance between the bases of the wings);
- inter-ocular distance;
- hind leg length (tibia, femur and tarsus);
- antennae.

Principle

We want to create a model that detects keypoints on insect images. Each trait will then be calculated using the coordinates of the corresponding points.



Measurements to be done and related keypoint placements

Each keypoint must be placed according to specific rules, which are described in the [How to place keypoints](#) section.

Setup of Label Studio

For annotation, we will use Label Studio, a free software tool that can be launched with pip, which is installed together with Python.

If you encounter any issues installing either software, please let me know so I can help.

Install python

Go to the website: <https://www.python.org/downloads/>.

We will use **Python 3.14** (<https://www.python.org/downloads/release/python-3143/>). At the bottom of the page, you will find the different installer versions, select the one that matches your device (Windows, MacOS, Linux, etc.).

Run the downloaded file.

During installation, **make sure to check “Add Python to PATH” before clicking “Install Now.”**

Install Label Studio

You do not need to visit a website to install Label Studio, but you can find documentation and help here: https://labelstud.io/guide/quick_start. If you encounter any issues, please let me know as soon as possible so that we can address them quickly.

1. Open a command interface (**Windows** : Windows+R, then write “cmd”; **MacOS** : Command (⌘) + Spacebar to open Spotlight, then write “ter”, **Linux** : Press Ctrl + Alt + T, or search for “Terminal” in your application menu).
2. Type this command : **`python -m pip install --upgrade pip`** in order to have the latest version of pip.
3. Type : **`pip install label-studio`** . It should install the 1.23.0 or more version.
If the message “pip: command not found” appears, it may be because Python was not added to PATH during installation. In that case, run the Python installer again. Otherwise, this guide may help:
<https://builtin.com/software-engineering-perspectives/pip-command-not-found>.
4. Type this command : **`pip install --upgrade django-environ`**
5. When the installation is done, type **`label-studio start`** in the command interface. Then wait for the window to open, or open Label Studio at <http://localhost:8080>.
It may have some dependency errors (package not in the right version). In this case, you can try to run this command : `pip install --upgrade [name of the package]`

6. You will have to **create an account**, by using an email and a password. **Please write down your email and password ; you won't be able to reset your password later on.** See the part [Issues](#) if it happens.
7. Click **Create** to create a project. Set a title, a description, and **go to the Labeling Setup section**, you will be able to add data later on.
8. Select **Custom Template**, at the end of the list.
9. Copy the text in the file [template.txt](#) that I must have sent you (or see in the [annexes](#) part, or in my github [here](#)), and paste it in the **code section** (next to Visual), then click **Save** (see below the expected Preview).
10. Now you can import as many pictures as you want ! (But upload gradually, do not import 1000 pictures at one time 🤔)

The preview should look like this (scroll to see the full interface) :



Head	Eyes	Thorax	Abdomen
head-top 1	left-eye 5	thorax-left 7	body-left 0
head-left 2	right-eye 6	thorax-right 8	body-right q
head-right 3		thorax-bottom 9	body-tip w
neck 4			

Left Antenna	Left forewing	Left hindwing	Left hind leg
left-antenna-0 e	left-forewing-base s	left-hindwing-base z	left-leg-0 b
left-antenna-1 t	left-forewing-tip d	left-hindwing-tip x	left-leg-1 y
left-antenna-2 a	left-forewing-front f	left-hindwing-front c	left-leg-2 i
	left-forewing-rear g	left-hindwing-rear v	left-leg-3 o

Right Antenna	Right forewing	Right hindwing	Right hind leg
right-antenna-0 p	right-forewing-base i	right-hindwing-base	right-leg-0
right-antenna-1 j	right-forewing-tip n	right-hindwing-tip	right-leg-1
right-antenna-2 k	right-forewing-front m	right-hindwing-front	right-leg-2
	right-forewing-rear	right-hindwing-rear	right-leg-3

Enter any remarks...

0 characters · 0 / 1 submissions Press Shift + Enter to Add + Add

Annotate the images

Get some images

We will primarily transfer the data using USB keys. However, if this is not possible, I can provide a link to download the data online. As the platform has limited storage capacity, I won't use this method too often.

Please **do not rename the images** after getting them, as I will use the original file names to retrieve the associated metadata.

How to use Label Studio

Start Label Studio

To start Label studio, go in your command interface (**Windows** : Windows+R, then write "cmd"; **MacOS** : Command (\⌘) + Spacebar to open Spotlight, then write "ter"; **Linux** : Press Ctrl + Alt + T, or search for "Terminal" in your application menu), and write ***label-studio start***.

Import pictures

From the **main menu**, select your project. I advise you to create a new project for each bundle of pictures you will receive.

In the project page, click **Import**. You can either select files directly from your computer or drag and drop images into the import area.

Once done, you should be able to **click on any image to annotate it** !

Annotate pictures

There is the picture on the top, labels under. You can zoom by holding Ctrl (Windows) or Cmd (macOS) and scrolling the mouse wheel. Click once on a label or press the binded key to select a label, then click on the image to place the point.

At the bottom, there is a **text area** where you should write anything unusual or uncertain about the annotation. For example, if there is a wing missing or if the eyes are not clearly visible. Try to use **short keywords** (left wing missing, eyes not visible, etc.), as I may filter annotations using those.

To place a keypoint, select a label, and click at the right place of the image. You can drag a point already placed. To delete a point, you have to select it by clicking it, then you can press the red trash button (in the right side of the screen, there is a menu listing the selected elements).

You also can adjust the size of the dots in **Settings -> Labeling Interface -> Configure settings**.

Saving your work

I encourage you to regularly **save** your annotations by clicking “**export**” on the **project menu** (select **JSON** format). This file will allow you to recover your work if there are any issues.

How to place keypoints

Before annotating, **rotate** each image (using the button on the right side of the interface) so that the **head is at the top**, the right side of the insect is at your right and the left side of the insect is at your left.

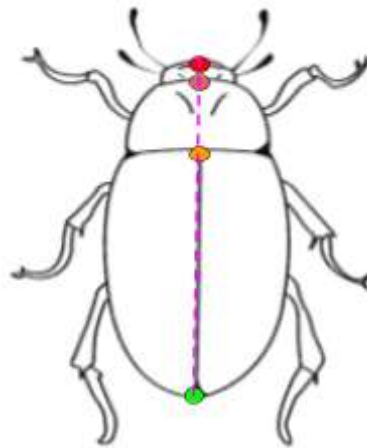
As I am not myself a professional in insect annotation, if you find any annotation errors on the next illustrations, please let me know 😊

Head, thorax and abdomen lengths

Place the following points:

- **head-top** at the very top of the insect;
- **neck** between the head and the thorax;
- **thorax-bottom** between the thorax and the abdomen;
- **abdomen-tip** at the very bottom of the insect.

These four points should be aligned, unless the insect's body is curved.



Example of middle-body keypoint placements

Head, thorax and abdomen widths

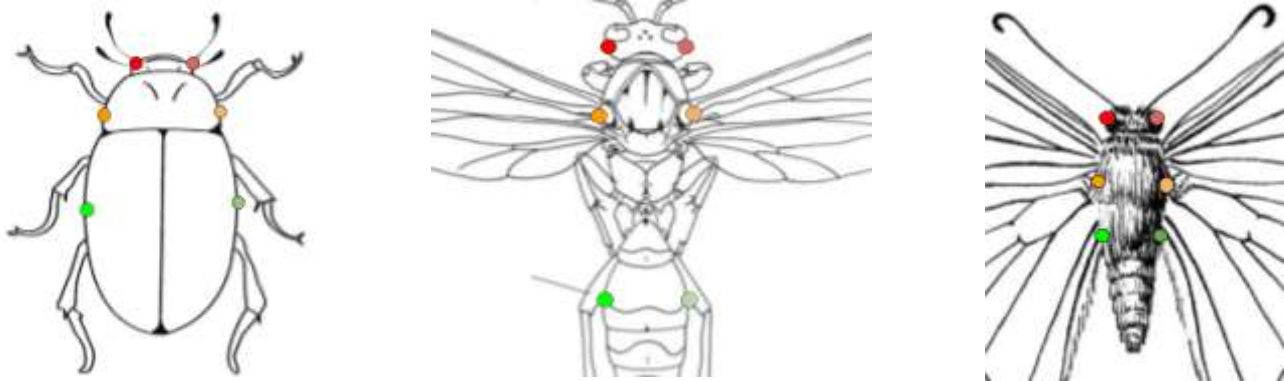
Place **head-left** and **head-right** on the head so that the distance between them is maximized.

Do the same for **body-left** and **body-right** on the abdomen.

For the **thorax**, place the keypoints on the widest part. The exact position depends on the species.

Examples:

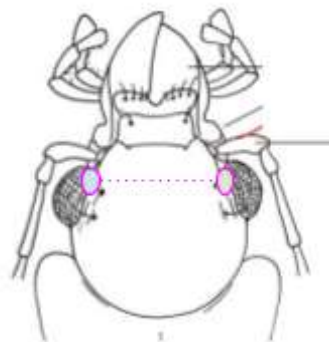
- Beetles: the widest part is usually the prothorax;
- Lepidoptera: the widest part is usually the mesothorax.



Examples of side-body keypoint placements

Eyes

If the eyes are visible, place **eye-left** and **eye-right** directly on the eyes so that the distance between them is **as small as possible**.



Example of eye keypoint placements

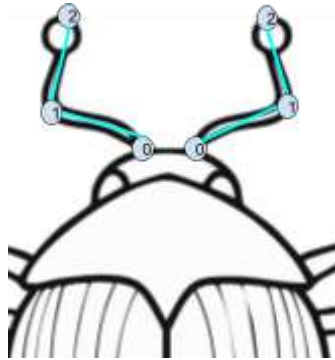
Antennae

For each antenna, place three points:

- **left-antenna-0** and **right-antenna-0**: base of the antenna on the head;
- **left-antenna-2** and **right-antenna-2**: tip of the antenna;
- **left-antenna-1** and **right-antenna-1**: middle point.

The **middle point** is used to represent bends in the antenna. Many times the antennae are curved and are not in exactly two segments. You should try to maximize the fit using this middle point anyway.

If an antenna is missing, do not place its keypoints.



Example of antenna keypoint placements

Wings (fore & hind)

For each wing, place four points:

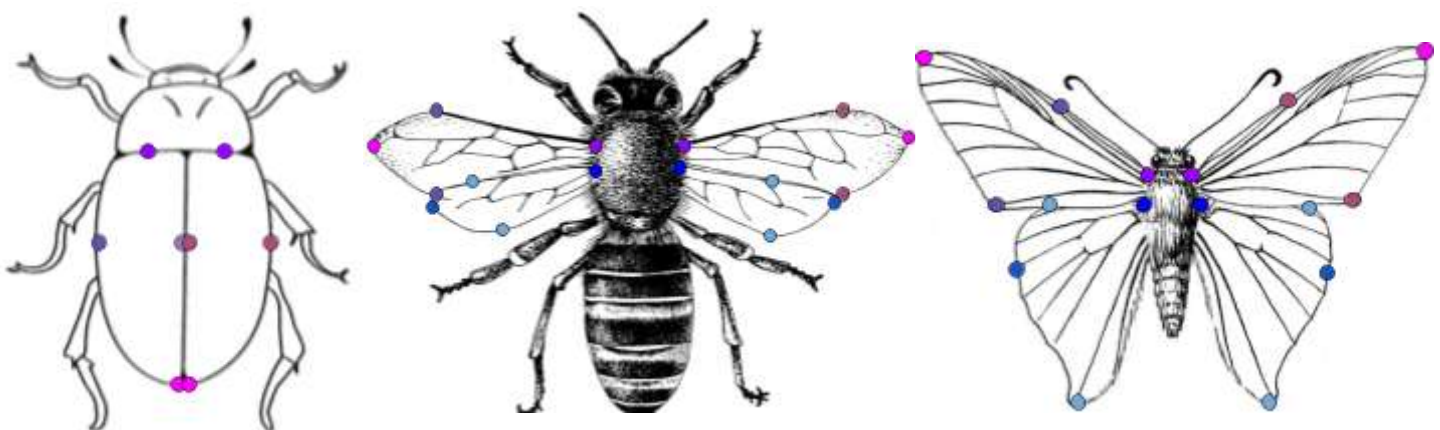
- **base**: where the wing attaches to the body;
- **tip**: the point farthest from the base;
- **front**: the front edge of the wing;
- **rear**: the rear edge of the wing.

The front and rear points should be positioned so that the distance between them is **as large as possible**.

Special cases:

- Beetles: if wings are folded, only place points on the elytra (forewing);
- If the insect has only one pair of wings, place only forewing points.

If wings are absent, do not place these points.



Examples of wing keypoint placements

Hind legs

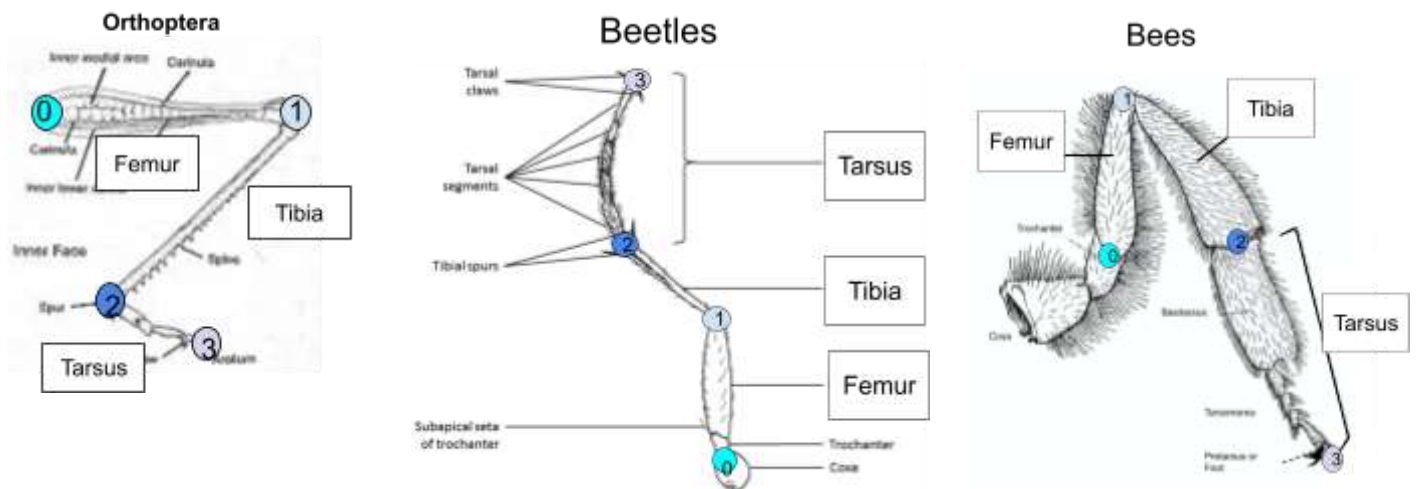
For each hind leg, place four points :

- **leg-left/right-0**: intersection with the main body;
- **leg-left/right-1**: joint between femur and tibia;
- **leg-left/right-2**: joint between tibia and tarsus.
- **leg-left/right-3**: end of the tarsus.

If the legs are hidden (e.g., behind wings), do not place the points.

If they are partially visible, place only the visible keypoints.

Because hind legs vary across species, here are some examples :



Examples of leg keypoint placements

Exporting data

Once you have finished annotating your set of images, **export** your saves into JSON format (**Export** button in the project menu), and send me the files at :

tom.bellivier@agroparistech.fr.

You will not need to send the images back to me, as the file names are included in the exported file. I will use these to link your annotations to the corresponding images.

Issues

If you have any issues with the software during installation or use of Label Studio, please let me know as soon as possible so that we can address them quickly.

If you have lost access to your account, please create a new one using a different email address. If your projects have disappeared but you have exported them frequently, you can restore your work by **re-importing your images and the JSON file** you previously exported.

You can also reach me by phone (whatsapp) : **+33 6 95 64 15 51**.

Final words

Good luck !

And in order to motivate you : I will bring a croissant to the person that annotates the more images each week 😊

Examples

Same as before, if you find any errors in the annotation below, let me know !





Head

- head-top 1
- head-left 2
- head-right 3
- neck 4

Eyes

- left-eye 5
- right-eye 6

Thorax

- thorax-left 7
- thorax-right 8
- thorax-bottom 9

Abdomen

- body-left 0
- body-right q
- body-tip w

Left Antenna

- left-antenna-0 e
- left-antenna-1 t
- left-antenna-2 a

Left forewing

- left-forewing-base s
- left-forewing-tip d
- left-forewing-front f
- left-forewing-rear g

Left hindwing

- left-hindwing-base z
- left-hindwing-tip x
- left-hindwing-front c
- left-hindwing-rear v

Left hind leg

- left-leg-0 b
- left-leg-1 y
- left-leg-2 i
- left-leg-3 o

Right Antenna

- right-antenna-0 p
- right-antenna-1 j
- right-antenna-2 k

Right forewing

- right-forewing-base l
- right-forewing-tip n
- right-forewing-front m
- right-forewing-rear

Right hindwing

- right-hindwing-base
- right-hindwing-tip
- right-hindwing-front
- right-hindwing-rear

Right hind leg

- right-leg-0
- right-leg-1
- right-leg-2
- right-leg-3

Annexes

Label studio template to copy-paste

Or copy the text at this address if it doesn't work:

https://github.com/TomBellivier/CV-for-GRIT/blob/main/label_studio_template.txt

```
<View>
```

```
  <View style="display: flex; flex-direction: column; justify-content: space-between; gap: 20px; padding: 15px; border-radius: 5px; margin-bottom: 10px;">
```

```
    <Image name="img-1" value="$img" zoom="true" zoomControl="true" rotateControl="true"/>
```

```
    <KeyPointLabels name="kp" toName="img-1" strokeWidth="4" maxUsages="1" unique="true" showInLine="false" opacity="0.5">
```

```
      <View style="display: flex; flex-direction: column; justify-content: space-between; gap: 5px; padding: 5px; border-radius: 5px; margin-bottom: 5px;" background="#ffa39e">
```

```
        <View style="display: flex; flex-direction: row; justify-content: space-between; gap: 5px; box-shadow: 2px 2px 5px #999; padding: 15px; border-radius: 5px; margin-bottom: 5px;">
```

```
          <View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
```

```
            <Header value="Head" style="font-size: 14px; border-bottom: 1px solid #eee; padding-bottom: 5px;">
```

```
              <View style="display: flex; flex-direction: column; justify-content: space-between;">
```

```
                <Label value="head-top" background="#ff0000"/>
```

```
                <Label value="head-left" background="#e1004b"/>
```

```
                <Label value="head-right" background="#e14b00"/>
```

```
                <Label value="neck" background="#c80000"/>
```

```
              </View>
```

```
            </View>
```

```
          <View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
```

```
            <Header value="Eyes" style="font-size: 14px; border-bottom: 1px solid #eee; padding-bottom: 5px;">
```

```
              <View style="display: flex; flex-direction: column; justify-content: space-between;">
```

```
                <Label value="left-eye" background="#c8c8dc"/>
```

```
                <Label value="right-eye" background="#c8dcc8"/>
```

```
              </View>
```

```
            </View>
```

```
          <View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
```

```
            <Header value="Thorax" style="font-size: 14px; border-bottom: 1px solid #eee; padding-bottom: 5px;">
```

```
              <View style="display: flex; flex-direction: column; justify-content: space-between;">
```

```
                <Label value="thorax-left" background="#ff9632"/>
```

```
                <Label value="thorax-right" background="#ffc800"/>
```

```

    <Label value="thorax-bottom" background="#ffff00"/>
  </View>
</View>

<View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
  <Header value="Abdomen" style="font-size: 14px; border-bottom: 1px solid #eee;
padding-bottom: 5px;">
    <View style="display: flex; flex-direction: column; justify-content: space-between;">
      <Label value="body-left" background="#64c864"/>
      <Label value="body-right" background="#64ff00"/>
      <Label value="body-tip" background="#00ff00"/>
    </View>
  </View>

</View>

<View style="display: flex; flex-direction: row; justify-content: space-between; gap:
20px; box-shadow: 2px 2px 5px #999; padding: 15px; border-radius: 5px; margin-bottom:
10px;">

  <View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
    <Header value="Left Antenna" style="font-size: 14px; border-bottom: 1px solid #eee;
padding-bottom: 5px;">
      <View style="display: flex; flex-direction: column; justify-content: space-between;">
        <Label value="left-antenna-0" background="#00ffff"/>
        <Label value="left-antenna-1" background="#96ffff"/>
        <Label value="left-antenna-2" background="#c8ffff"/>
      </View>
    </View>

    <View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
      <Header value="Left forewing" style="font-size: 14px; border-bottom: 1px solid #eee;
padding-bottom: 5px;">
        <View style="display: flex; flex-direction: column; justify-content: space-between;">
          <Label value="left-forewing-base" background="#c800ff"/>
          <Label value="left-forewing-tip" background="#c80096"/>
          <Label value="left-forewing-front" background="#ff00c8"/>
          <Label value="left-forewing-rear" background="#9600af"/>
        </View>
      </View>

      <View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
        <Header value="Left hindwing" style="font-size: 14px; border-bottom: 1px solid #eee;
padding-bottom: 5px;">
          <View style="display: flex; flex-direction: column; justify-content: space-between;">
            <Label value="left-hindwing-base" background="#646496"/>
            <Label value="left-hindwing-tip" background="#6464ff"/>
            <Label value="left-hindwing-front" background="#9664c8"/>
            <Label value="left-hindwing-rear" background="#3264c8"/>
          </View>
        </View>
      </View>
    </View>
  </View>
</View>

```

```

    </View>
  </View>

  <View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
    <Header value="Left hind leg" style="font-size: 14px; border-bottom: 1px solid #eee;
padding-bottom: 5px;" />
    <View style="display: flex; flex-direction: column; justify-content: space-between;">
      <Label value="left-leg-0" background="#00c8ff" />
      <Label value="left-leg-1" background="#7dc8ff" />
      <Label value="left-leg-2" background="#afc8ff" />
      <Label value="left-leg-3" background="#ffc8ff" />
    </View>
  </View>

</View>

<View style="display: flex; flex-direction: row; justify-content: space-between; gap:
20px; box-shadow: 2px 2px 5px #999; padding: 15px; border-radius: 5px; margin-bottom:
10px;">

  <View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
    <Header value="Right Antenna" style="font-size: 14px; border-bottom: 1px solid
#eee; padding-bottom: 5px;" />
    <View style="display: flex; flex-direction: column; justify-content: space-between;">
      <Label value="right-antenna-0" background="#00ffc8" />
      <Label value="right-antenna-1" background="#96ffc8" />
      <Label value="right-antenna-2" background="#c8ffc8" />
    </View>
  </View>

  <View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
    <Header value="Right forewing" style="font-size: 14px; border-bottom: 1px solid
#eee; padding-bottom: 5px;" />
    <View style="display: flex; flex-direction: column; justify-content: space-between;">
      <Label value="right-forewing-base" background="#c83296" />
      <Label value="right-forewing-tip" background="#c89696" />
      <Label value="right-forewing-front" background="#ff6496" />
      <Label value="right-forewing-rear" background="#966496" />
    </View>
  </View>

  <View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
    <Header value="Right hindwing" style="font-size: 14px; border-bottom: 1px solid
#eee; padding-bottom: 5px;" />
    <View style="display: flex; flex-direction: column; justify-content: space-between;">
      <Label value="right-hindwing-base" background="#9696ff" />
      <Label value="right-hindwing-tip" background="#969696" />
      <Label value="right-hindwing-front" background="#c896c8" />
    </View>
  </View>

```

```

        <Label value="right-hindwing-rear" background="#6496c8"/>
      </View>
    </View>

    <View style="flex: 1; border-left: 1px solid #eee; padding-left: 15px;">
      <Header value="Right hind leg" style="font-size: 14px; border-bottom: 1px solid
#eee; padding-bottom: 5px;"/>
      <View style="display: flex; flex-direction: column; justify-content: space-between;">
        <Label value="right-leg-0" background="#00c8c8"/>
        <Label value="right-leg-1" background="#7dc8c8"/>
        <Label value="right-leg-2" background="#afc8c8"/>
        <Label value="right-leg-3" background="#ffc8c8"/>
      </View>
    </View>

    </View>
  </View>
</KeyPointLabels>
  <TextArea name="remarks" toName="img-1" placeholder="Enter any remarks..."
rows="5" maxSubmissions="1"/>

</View>

</View>

```